MIL-D-4562(USAF)
10 October 1952
Superseding
No. 50041-A
31 October 1941

MILITARY SPECIFICATION

DIES, THREADING, DIE HEAD

1. SCOPE

1.1 Scope. - This specification covers self-opening dies complete with adjustable and removable die chasers of such design and construction as to operate on hand turret lathes for cutting tapered or straight threads.

1.2 Classification. - Dies shall be of the following types, classes, sizes and threads.

Type I - Taper thread type

Class NPT - for taper threads and sizes conforming to Handbook H28
Class ANPT - for taper threads and sizes conforming to Specification MIL-P-7105

Type II - Straight thread type - for straight threads, sizes and classes conforming to Handbook H28

2. APPLICABLE SPECIFICATIONS, STANDARDS, DRAWINGS AND PUBLICATIONS

2.1 The following specifications and standards of the issue in effect on date of invitation for bids, form a part of this specification:

SPECIFICATIONS

FEDERAL

NN-B-601	Boxes; Wood-Cleated-Plywood, For Domestic Shipment
NN-B-621	Boxes; Wood, Nailed and Lock-Corner
LLL-B-631	Boxes; Fiber Corrugated (For Domestic Shipment)
LLL-B-636	Boxes; Fiber, Solid (For Domestic Shipment)

MILITARY

JAN-P-105	Packaging and Packing for Overseas Shipment; Boxes, Wood, Cleated Plywood
JAN-P-106	Packaging and Packing for Overseas Shipment; Boxes, Wood, Nailed
MIL-P-116	Preservation, Methods of
JAN-P-120	Packaging and Packing for Overseas Shipment, Cartons, Folding Paperboard
JAN-P-133	Packaging and Packing for Overseas Shipment, Boxes, Set-Up Paperboard

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MIL-P-7105	Pipe-Threads; Taper, Aeronautical National Form, Symbol ANPT
MIL-S-7742	Screw Threads; Standard, Aeronautical
MIL-M-7911	Marking, Identification of Aeronautical Equipment, Assemblies, and Parts

STANDARDS

MILITARY

MIL-STD-129 Marking of Shipments

(Copies of specifications, standards, and drawings required by contractors in connection with specific procurement functions should be obtained from the procuring agency or as directed by the contracting officer.)

2.2 Other publications. - The following publication, of the issue in effect on date of invitation for bids, unless otherwise stated, forms a part of this specification:

National Bureau of Standards Publication

Handbook H28 Screw-Thread Standards for Federal Services and 1950 Supplement thereto.

(Copies of Handbook H28 and 1950 Supplement thereto may be obtained upon application, accompanied by money order, coupon, or cash to the Superintendent of Documents, Government Printing Office, Washington 25, D. C.)

3. REQUIREMENTS

3.1 Materials. - Materials used in the construction of these die heads shall be of high-grade quality throughout, and entirely suitable for the purpose.

3.2 General. -

3.2.1 Fixtures for regrinding the thread chasers shall be furnished when specified. (See 6.3)

3.2.2 Provision shall be made in the die head to securely hold the thread chasers in the proper positions for threading.

3.2.3 The thread chasers shall be readily and easily removable.

3.2.4 Provision shall be made for adjusting the thread chasers to obtain the proper pitch diameter.

3.2.5 Thread chasers shall be manufactured of high-grade, high-speed tool steel, suitably hardened and tempered and shall be free from all defects that might effect their accuracy and serviceability.

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3.2.6 The thread chasers shall be capable of threading to within the equal of one thread from a shoulder without seriously marring the face of the shoulder.

3.2.7 The throat of the straight thread chasers shall be chamfered not to exceed two threads and the throat of the taper thread chasers shall be chamfered not to exceed one thread.

3.2.8 A set of chasers shall consist of the number of chasers required for use in a die head to cut one size of thread.

3.2.9 The shank of die heads for use with straight thread chasers shall be provided with a hole of sufficient clearance for the maximum capacity of the range of threads and diameters, and the maximum thread length as specified. (See 6.3)

3.2.10 Provisions shall be made in the die head to automatically open the chasers at a pre-determined length of thread. A manually operated device for opening the chasers by hand shall also be provided.

3.2.11 The length and diameter of the shank, the maximum outside diameter of the die head, and the material to be threaded shall be as specified. (See 6.3)

3.2.12 The die head shall be substantially in accordance with Figure 1.

3.3 Tolerances. -

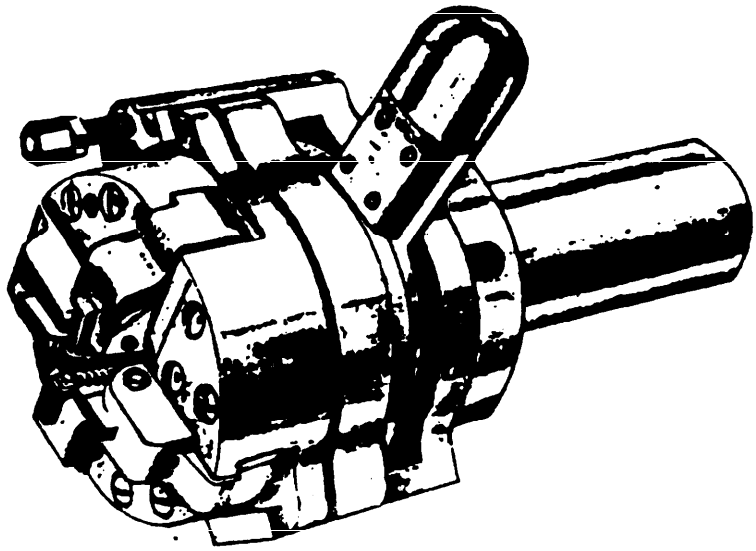
3.3.1 Taper thread chasers. - The chasers for taper pipe threads, when installed in their die head, shall be of sufficient accuracy to produce threads as follows:

3.3.1.1 Class NPT. - Taper threads conforming to Handbook H28.

3.3.1.2 Class ANPT. - Taper threads conforming to Specification MIL-P-7105

3.3.2 Straight thread chasers. - Straight thread chasers, when installed in their die head, shall be capable of cutting a clean, smooth thread, full profile, and the combined error in half angle and lead of the threads cut shall not exceed half of the pitch diameter tolerances as specified in Handbook H28 and 1950 Supplement thereto, for the applicable class.

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Die Head Threading Die

Figure 1

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3.4 Identification of product. - The die shall be marked for identification in accordance with Specification MIL-M-7911. The name plate shall contain the following:

Die, Threading, Die Head
Specification MIL-D-4562
Type_____
Class_____
Thread_____
Stock No._____
Mfr's Part No._____
Serial No._____
Contract or Order No.___
Mfr's Name or Trademark
U. S. Property

3.5 Chaser marking. - Each individual chaser of each set shall be marked with the thread size, symbol and a number to indicate the position of chaser in the die head.

3.6 Workmanship. - Workmanship shall be in accordance with high-grade manufacturing practice covering this class of equipment.

4. SAMPLING, INSPECTION AND TEST PROCEDURES

4.1 Inspection. - The die heads and all material entering into the construction thereof, shall be subject to inspection by authorized Government inspectors. When inspection is conducted at the contractor's plant, tests shall be conducted by the contractor under the supervision of the authorized Government inspector.

4.2 Tests. - Each die head shall be tested under actual conditions using a set of chasers selected at random by the Government inspector from the chasers to be supplied on the order or contract. Cutting tests shall be conducted on the materials for which the chasers have been specified in the order or contract, and be capable of producing threads, as specified. (See 6.3)

4.3 Acceptance or approval of material in course of manufacture shall in no case be construed as a guarantee of the acceptance of the finished product.

4.4 Acceptance or rejection of die heads and chasers will be based on their ability to produce threads within the limits of this specification.

5. PREPARATION FOR DELIVERY

5.1 Application. - The packaging, packing and marking requirements specified herein apply only to direct purchases by or direct shipments to the Government.

5.2 Interior packaging. -

5.2.1 Chasers. - One set of chasers of each thread size shall be preserved in accordance with Specification MIL-P-116, Method IB and shall be packaged in containers conforming to Specification JAN-P-120 and JAN-P-133.

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5.2.2 Die heads. - The die heads shall be preserved in accordance with Specification MIL-P-116, Method IB and shall be packaged in containers conforming to Specification JAN-P-120 or JAN-P-133.

5.3 Exterior packing. -

5.3.1 Domestic shipment. - Unless otherwise specified, packing for domestic shipment shall be in boxes in accordance with Specifications MN-B-601, MN-B-621, LLL-B-631 or LLL-B-636.

5.3.2 Overseas shipment. - When export shipment is specified, the die heads and chasers shall be packed in boxes in accordance with Specification JAN-P-105 or JAN-P-106.

5.4 Marking of shipments. - Interior packages and exterior shipping containers shall be marked in accordance with Standard MIL-STD-129. The nomenclature shall be as follows:

Die, Threading, Die Head
Type *
Class *
Thread Range or Size of Thread
Mfr's. Part No. *
Specification MIL-D-4562

*Applicable data to be entered by contractor.

6. NOTES

6.1 Intended use. - Die heads, complete with adjustable and removable chasers, covered by this specification, are intended for use on hand turret lathes for cutting tapered or straight threads within the range of threads and diameters specified by the Procuring Agency.

6.2 Illustrations. - The illustration shown herein is for the convenience of identification, requisitioning, purchasing, and identification and is not intended to preclude the purchase of die heads or chasers which are otherwise in accordance with the requirements of this specification.

6.3 Ordering data. - Procurement documents should specify the following:

a. Title, number and date of this specification.
b. Type, class and size of threads required (see 1.2).
c. If fixtures for regrinding are required (see 3.2.1).
d. Length and diameter of the shank (see 3.2.11).
e. Maximum outside diameter of the die head (see 3.2.11).
f. Material to be threaded (see 3.2.11).
g. When overseas shipment is required (see 5.3.1 and 5.3.2).

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5. PROBLEM AREAS

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6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
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